

Addala Raed *Software Engineering Student*

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PROFESSIONAL EXPERIENCE

Distributed Computing Intern, Imaginario AI (Remote) ☑ 06/2025 – 08/2025 | Philadelphia, USA
Imaginario AI is a startup developing AI-powered tools for automated video understanding and content creation, leveraging multimodal AI embedding and technology.

- Designed and implemented a distributed video processing system with **context-aware chunking** for efficient parallel analysis.
- Applied a **MapReduce**-inspired architecture to distribute embedding and processing tasks across multiple compute nodes.
- Improved scalability and performance of video analysis pipelines by enabling parallel processing and recombination of results.

Junior Software Engineer, BySur (Remote) ☑ 11/2024 – 04/2025 | Paris, France
BySur provides digital solutions tailored for the insurance industry. As a working student, I contributed to the development of backend services and Retrieval-Augmented Generation (RAG) systems.

- Developed a **RAG** pipeline using **FastAPI** and **locally hosted CUDA-accelerated LLMs**, enabling intelligent document querying and summarization for the insurance domain.
- Built and maintained **fully containerized services** with **Docker**.
- Contributed to modular, maintainable backend development using **NestJS** with a focus on clean architecture and system integration.
- Worked across the stack, supporting **ReactJS-based frontend** development in collaboration with UI teams when necessary.
- Leveraged **Python**, **TypeScript**, and **Docker** as core technologies across multiple projects in a fast-paced, agile environment.

Junior Software Engineer, Metadrift ☑ 03/2024 – 10/2024 | Tunis, Tunisia
Worked primarily as a backend developer on a custom ERP system for our client SilverPharma, tailored to automate financial operations and stock management. Involved across the full development cycle from design to deployment.

- Designed and developed scalable and modular backend services using Nest.js and TypeScript.
- Developed tailored ERP features to streamline client workflows and reduce manual errors.
- Optimised SQL queries and database design for faster reporting.
- Automated key business processes, improving operational efficiency.
- Created modular components for future ERP expansion.
- Managed Docker environments and CI/CD pipelines for consistent delivery.

Tech Stack: NestJS, TypeScript, Docker, PostgreSQL, React.js.

Software Engineer Intern, Prime Speed Transport 01/2024 – 03/2024 | Dubai, UAE (Remote)
Prime Speed Transport is a public transportation company providing dynamic bus services, primarily for Dubai International Airport staff.

- Designed and implemented a dynamic logistics algorithm to automate bus allocation and routing, adapting to changing customer demands and significantly improving operational efficiency and service quality.
- Integrated the algorithm with Google Sheets to automate key management tasks, streamlining daily planning and increasing reliability.
- Gained hands-on experience solving real-world logistics challenges through code, applying skills in algorithm design, JavaScript, and operations research.

Mobile Developer Intern, MS Solutions ☑ 07/2023 – 09/2023 | Tunis, Tunisia
Developed a hotel booking mobile application in **Flutter 3** with Material 3 design guidelines, using **GetX** for state management, **Retrofit** for API integration, and **Firebase** and **Supabase** for authentication, data storage, and real-time access.

- Applied Flutter's widget-based architecture and declarative design for clean, modular, and responsive UI development.
- Applied clean architecture principles and design patterns to ensure maintainable, scalable code.
- Collaborated in a development team, following version control workflows and clear technical communication practices.

PROJECTS

Fast Smart Contract System Extension for Parallel Cluster-Based PBFT 02/2025 – 06/2025
Blockchain Platform, End-of-Year Project

Under the supervision of Prof. Dr. Sofiane Ouni and based on his research on Random Cluster PBFT (RC-PBFT) published as an article in the Springer Cluster Computing Journal ☑, my colleague and I developed an advanced smart contract system extension. This extension introduces a **synchronisation** module designed to *harness the parallelism of RC-PBFT* and a runtime environment. Our goal was to *adapt* blockchain infrastructure for high-throughput use cases, particularly in IoT. We focused on **minimising inter-node conflicts through enhanced synchronisation**. The system was developed in **Go**.

SmartShield – Real-Time Intrusion Detection System, 09/2024 – 12/2024
2nd Place – IEEE TSYP12 Computer Society Challenge ☑

Worked in a 7-person team to develop a modular, containerised intrusion detection platform leveraging AI-driven behavioural analysis to detect and classify cyber threats from network logs in real-time.

- Designed a high-throughput, low-latency log processing pipeline using **RabbitMQ**, **Zeek**, **OpenArgus**, and **Elasticsearch**.

- Implemented **anomaly detection** and **multi-class classification models** to improve threat identification.
 - Integrated an **LLM-powered agent** for generating explainable security reports.
 - Built an interactive dashboard for **real-time monitoring and response**.
 - Ensured scalability and extensibility via **Dockerized microservices architecture**, supporting future integrations (e.g., sandboxing, **YARA rules**)
- Technologies:** Docker, Python, TypeScript, NestJS, RabbitMQ, OpenArgus, Zeek, and Elasticsearch.

Microservice-based Live Streaming Platform, End-Of-Year Project [↗](#)

02/2024 – 06/2024

- Technologies:** Kong, Elixir (Phoenix), SvelteKit, Membrane, Docker
- Designed and built a **fault-tolerant, highly available** live streaming platform in a three-person team using a microservice architecture.
 - Developed a resilient **Elixir/Phoenix** API leveraging BEAM VM **concurrency** and **fault tolerance** for real-time streaming workloads.
 - Implemented **Kong** API Gateway for secure routing, load balancing, request verification, and asymmetric JWT authentication.
 - Engineered an end-to-end streaming pipeline: **RTMP** HD video **ingestion**, **real-time transcoding** (AAC, H.264), and **adaptive HLS delivery**.
 - Built a low-latency, bidirectional live chat service using WebSockets for audience engagement.
 - Delivered an intuitive, responsive frontend with **SvelteKit**, ensuring a seamless user experience.
 - Containerised services with **Docker** for consistent deployment and scalability.
- For More Details, check our Project Report [↗](#) .

SKILLS

• C/C++	• Python	• Typescript	• Elixir
• Go	• Java	• Node.js	• Nest.js
• React.js	• SQL	• PostgreSQL	• OOP
• Design Patterns	• Data Engineering	• Linux	• Docker
• Kubernetes	• RabbitMQ	• Kafka	• PyTorch

LANGUAGES

English — Fluent	French — Proficient
Arabic — Native/Bilingual	German — Basic

EDUCATION

Software Engineering Degree, <i>National Institute of Applied Sciences and Technology - INSAT</i> ↗	2021 – 2026 Tunis, Tunisia
Baccalaureate in Mathematics with Honors, <i>Pioneer High School Medenine</i>	2020 – 2021 Medenine, Tunisia

EXTRA ACTIVITES

Competitions Platform System Administrator, <i>INSAT ACM Student Chapter</i> Collaborated with a team in organising national-level competitive programming contests, focusing on system administration and setup and management of the competition infrastructure, including network setup and platform design using DomJudge . We ensured high availability and scalability by containerising components and orchestrating with Kubernetes.	12/2022 – present Tunis, Tunisia
Problem Setter and Contest Tester in Competitive Programming Contests, <i>INSAT ACM Student Chapter</i> • Designed and developed challenging, original problems for competitive programming contests. • Created solutions with thorough correctness proofs and conducted algorithmic complexity analysis. • Developed comprehensive test cases, including edge cases and stress tests, to validate solution robustness and performance. • Conducted full problem set testing to verify the suitability of the contest duration (5-hour format). • Volunteered at <i>INSAT Winter Cup</i> (Edition 6.0, 7.0) and <i>INSAT ACM CodeQuest</i> (Edition 1.0, 2.0).	12/2023 – present Tunis, Tunisia
INSAT ACM Student Chapter - Training Manager • Organized and supervised weekly Competitive Programming workshops and training competitions to prepare members for the TCPC (Tunisian Collegiate Programming Contest), the national ICPC qualifier. • Delivered select workshops and provided ongoing support during training sessions.	08/2023 – 06/2024 Tunis, Tunisia
Technical Activites Manager, AI National Summit (AINS) 2.0, <i>IEEE Computer Society INSAT Student Branch</i> • Led technical planning for IEEE CS INSAT SB's AINS 2.0, featuring talks, workshops, and competitions. • Defined summit themes, proposed and secured speakers, and coordinated with workshop instructors and competition juries. • Organised and managed a Data Science Competition ↗ and an AI Hackathon, ensuring technical rigour and smooth execution.	02/2024 – 06/2024 Sousse, Tunisia